Review of Models Proposed for dealing with Occlusion in Facial Expression Recognition

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***Abstract*—In recent years, facial research recognition has advanced to a point where it can reliably detect faces in controlled settings, but it still struggles with occlusion and in real world scenarios. This paper reviews different models proposed to deal with occlusion. This paper also comments on the various strategies to used to detect occlusion.**

This paper contains an analysis of various models proposed. Along side that, there is a short analysis of the common datasets that are used for training and testing these models.

This paper found that

1) **Models that combine both global and local features outperform those who do not.**
2) **No model attained a satisfactory score when tested on real occluded datasets.**
3) **Several models are constrained by their reliance on facial landmark detection models/algorithms.**

***Index Terms*—Computer Vision, Facial Expression Recognition, Occlusion Handling, Deep Learning**

I. Introduction

FACIAL Expression Recognition, or FER has undergone numerous advancements in recent years. We are now at a point where we have models that can perform FER with high degrees of accuracy in laboratory settings [1].

However, occlusion still poses a problem to reliable FER. Occlusion refers to the phenomenon which occurs when a portion of a 3D scene is not visible from the camera or another sensor [2]. Many real world applications will need to deal with occlusion in order to be of any use. Keeping this in mind, it is essential that researchers deal with the occlusion problem sufficiently.

Researchers have been trying to deal with occlusion in FER (OFER) for over 20 year now, with Bourel et al. publishing one of the earliest works on the topic [3]. Since then numerous models have been proposed to deal with the issue. There have been several reviews on methods to detect FER have been published and among the first to include methods that deal with occlusion was Bettadapura [4] in 2012. A much more comprehensive survey was conducted by Zhang et al. [5] in 2018.

II. Background

In this section, a discussion of the standard FER pipeline is presented. This emerged over decades of research related to facial expressions primarily by psychologists. In the 1990s and 2000s, when machine learning techniques were first applied, this pipeline solidified [14][15]. The pipeline is given as:

1) Face Detection
2) Preprocessing
3) Feature Extraction
4) Classification

A. Occlusion Scenarios

There are numerous scenarios in real world settings where the face is subject to varying degrees of occlusion. Face masks, glasses and caps [12][18] all can cause occlusion and lead to loss of effectiveness in FER models.

Pose can also contribute to occlusion. When someone turns their head, for example, several important features of the face get hidden, again causing the effectiveness to be hit [19].

Another scenario that leads to occlusion is termed ”self-occlusion”, where the person blocks an area of their face, either deliberately or subconsciously. For example, someone rubbing their eyes [19].

B. Why Occlusion Hurts FER

If pixels are simply missing, then that is information loss. This is very bad as it is, but occlusion is a step further. It causes large scale spatial changes in the face [20].

III. Datasets

A major challenge researchers face in this field is the availability of datasets. There exist several datasets of human faces. These can be categorized as either in-the-lab datasets or in-the-wild datasets. Examples of in-the-lab datasets include CK+ [6], MMI [7], Oulu-CASIA [8] and JAFFE [16]. Examples of in-the-wild datasets include RAF-DB [9], AffectNet [10], SFEW [11] and FER2013 [17].

One thing to note about all these datasets is that none of them contain images with partial occlusion. All of them are clear face images. To the best of my knowledge, there is only one dataset that has collected faces with organic occlusions, which is the FED-RO dataset [12].

RAF-DB contains 30,000 facial images annotated with basic or compound expressions by 40 trained human coders. The Extended Cohn-Kanade dataset (CK+) contains 593 video sequences recorded from 123 subjects. MMI dataset includes more than 30 subjects of both genders. There are 79 sequences

of each subject. Each begins and ends with neutral facial expression. Oulu-CASIA dataset contains six prototypic expressions from 80 people between 23 to 58 years old [12]. SFEW The Static Facial Expressions in the Wild (SFEW) dataset is created by selecting static frames from Acted Facial Expressions in the Wild (AFEW) [13] This dataset contains 95 subjects in total and covers unconstrained facial expressions, different head poses, ages, and occlusions. In total there are 663 well-labelled usable images. The Japanese Female Facial Expression (JAFFE) database contains 213 samples of posed expressions from 10 Japanese females. Each person has 3 to 4 images with each of six basic facial expressions and one image with a neutral expression.

IV. Models Reviewed

This paper has reviewed the below 9 models:

1) Patch-Gated CNN (PG-CNN) [21]
2) Patch-Based ACNN (pACNN) [12]
3) Global-Local-Based ACNN (gACNN) [12]
4) OADN [22]
5) WGAN for OFER [23]
6) WLS-RF [24]
7) Privileged Information CNN (PI-CNN) [25]
8) RGBT [26]
9) CT-DBN [27]

Out of these, PG-CNN [21], pACNN [12], gACNN [12], OADN [22] and WLS-RF [24] can be classified as models that break down the face into patches, determine how useful each patch is and then use this information for classification.

WGAN for OFER [23] and PI-CNN [25] uses a generative approach to tackle occlusion. They deal with occlusion by trying to recreate the face if there was no occlusion in the first place.

A. Patch-Gated CNN and Patch-Based ACNN

PG-CNN is based on the first 9 convolutional layers of VGG-16 [30]. This gives feature maps of dimensions 512x28x28 .The feature maps from VGG-16 is then fed through a Region Decomposition algorithm which was proposed in [29] where 68 land mark points are found. Out of these, 16 are selected to cover eyebrow corners, eye corners, nose bridge, tip of the nose, corners of the mouth. 8 more points were derived. 4 midpoints for the eyes and eyebrows were calculated and 4 midpoints were calculated of the cheeks. The 512x28x28 feature map is then cropped around these 24 landmarks to give 24 local feature maps of dimension 512x6x6. Then, each of these 24 local feature maps are processed independently in their own patch-gate units. In each PGU, the feature maps go through 2 convolutions keeping the dimensions same. This output is then split along 2 branches. The first branch does feature encoding. It takes the feature map and encodes it into a 64-dimensional vector (512x6x6 to 128x3x3 to 64) [21].

The second branch is what the authors call an "attention net", which gives how occluded that patch is. The attention net gives an output between 0 and 1 via a sigmoid and represents unobstructed-ness. The output of the attention net is a simple

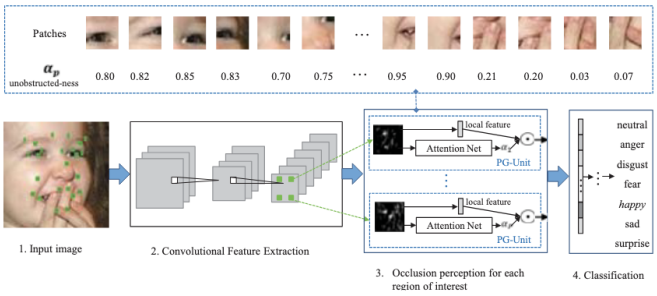

Fig. 1: Architecture of PG-CNN [21]

scalar. The final output of the PGU (Patch Gate Unit) is just the occlusion score times the feature vector. The weighted feature vectors from all 24 PGUs are then concatenated into a single large vector which is passed through 3 fully connected layers and finally a softmax classifier that outputs the probability for 7 different emotions, namely angry, disgust, fear, happy, sad, surprise, neutral. The architecture is represented in Figure 1 [21].

It is worth noting that this model was trained on synthetic occlusions. The occlusions added were constrained to the size range of 96x96 to 128x128 pixels. This range was chosen because smaller occlusions had little effect on the model [21].

Patch-Based ACNN has an architecture identical to PG-CNN with the sole difference being that PG-CNN has 3 fully connected layers and pACNN has 2. The similarity can be explained simply by realising that the authors of both [21] and [12] are the same and pACNN is a more refined version of PG-CNN [12].

B. Global-Local-Based ACNN

gACNN, as proposed in [12] is a further refinement of the PG-CNN architecture. While the PG-CNN [21] and pACNN [12] architectures are good at locating occlusions, what they sometimes miss is the larger picture. It is now enough to know that this part of the face has untrustworthy information, we also need to know the greater details of the face to place the untrustworthy patch in a broader context [12].

To solve this, the authors took the identical framework of the pACNN, but added a global gated unit (GGU) which is identical in function and structure to the PGU, but this works on the whole face. There is only one other notable difference between gACNN and pACNN, which is that instead of using the first 9 convolutional layers of VGG-16, they use the first 12, this is done to get slightly larger spatial dimensions for the feature map that they send to the GGU. Figure 2 shows the architecture of gACNN [12].

C. Occlusion Aware Deep Network (OADN)

This model uses ResNet-50 as its backbone, instead of VGG-16. It uses all the convolutional layers of ResNet50 [31], the only changes it does is removing the average pooling and fully connected layers and changing the stride of conv4_1 from 2 to 1 to get a larger feature map. This is because both

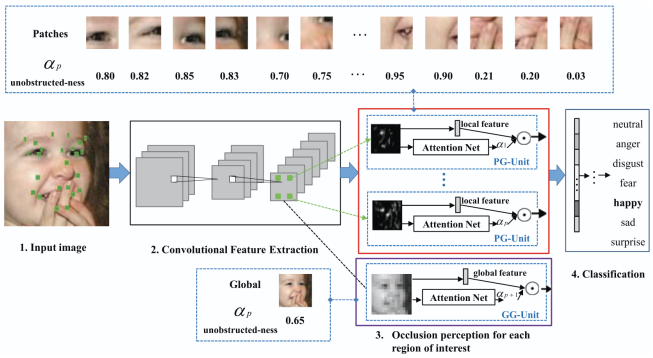

Fig. 2: Architecture of gACNN [12]

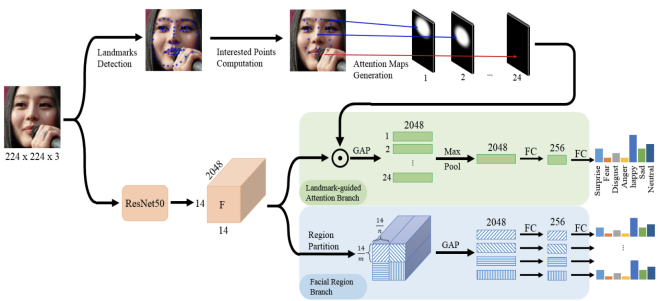

Fig. 3: Architecture of OADN [22]

branches need as detailed spatial information that they can get [22] .

After ResNet, The model splits into 2 branches. The first branch is the Landmark-guided Attention Branch or LAB. Firstly, a pretrained landmark detector, SAN [32] which is trained on the 300W dataset [33] extracts 68 facial landmarks. It outputs both the coordinates and a confidence score between 0 to 1, basically saying how confident the SAN is in its detection. From these 68, 24 points are selected. 16 are directly selected for covering eyes, eyebrows, nose and mouth, 8 are recomputed for cheeks and the areas between eyes and eyebrows. 2 things are important to note here. First, there seems to be some discrepancy in how many points are taken from the 68. Though they first claim that 16 are directly selected and 8 are recomputed, in a later section they claim that 18 are directly selected. Secondly, we are not told how the recomputed points are exactly recomputed. Though, considering how they cite the gACNN papers [12], it is likely that they use the very same method, but it is not explicitly mentioned. For the recomputed points, the confidence score is assigned of the parent point with the lower score. Say a point is recomputed from points with confidence scores of 0.76 and 0.59, the recomputed point will have a score of 0.59 [22].

The model applies a confidence threshold (denoted by T) which the authors set to 0.6. Any point with confidence score below T is considered to be occluded and hence discarded. Now for the interest points which are not occluded, a 2D gaussian heat map is applied to the point where the landmark is situated. Think of the landmark as a bulb. The landmark will have the brightest light, and the brightness of the light will fade as you go farther away from the bulb. The gaussian heat map tells how bright a point will be. Mathematically, it is expressed as:

$$G(x, y) = \exp\left(-\frac{(x - x_0)^2 + (y - y_0)^2}{2\sigma^2}\right)$$

Where σ controls how wide the glow spreads. For occluded points, the heat map produced has all zeros [22].

The heatmaps are then downsampled to be of 14x14 resolution using linear interpolation, so as to match the resolution of the backbone's feature map. Linear interpolation is used here instead of any other method because it compresses the heatmap while preserving the general shape of the gaussian. The backbone produces a feature map of dimensions 2048x14x14, so the heat map applied is applied across all 2048 channels. Now, we have 24 such heatmaps, all of which are applied on the same feature map to get 24 separate copies, all masked differently. Each of these 2048x14x14 feature maps is then global average pooled into a 2048D vector. All of these vectors represent what the network "sees" around a landmark. Then these 24 vectors are max pooled across the 24 landmarks, that is, for each of the 2048 dimensions, the maximum value across the 24 vectors is taken, fusing the information from all the regions into a single 2048D vector. This 2048D vector is then reduced to a 256D vector using a fully connected layer and is then fed into a softmax classifier which gives the probability distribution over the 7 classes, namely angry, disgust, fear, happy, sad, surprise, neutral. This branch is trained according to cross-entropy loss [22].

The second part of the OADN is the facial region branch. This branch takes the same 2048x14x14 feature map and divides the 14x14 spatial grid into K non-overlapping blocks. The authors found that $K = 4$ gives the best results. Each block, independently, is reduced to a 2048D vector, which is passed through its own FC layer, which reduces it to a 256D vector, which is passed on to a softmax layer which predicts the expression category. Each block's classifier is trained independently using cross entropy loss and the total loss is the sum of individual losses across all 4 regions [22].

There is some ambiguity as to how the model actually predicts a single class. The authors have told that the final output is the average of the LAB output and the FRB output. The ambiguity arises on how this operation is exactly done? Is it

$$\frac{LAB + average\ of\ 4\ regions}{2}$$

Or,

$$\frac{LAB + region1 + region2 + region3 + region4}{5}$$

It is likely that the first interpretation is used since the paper itself admits that the LAB is more effective on its own than the FRB on its own [22].

OADN uses 2 branches because the LAB is only good as the points it selects. So it is plausible that the image has a person wearing a mask, making the landmark detection algorithm to perform poorly, leading to the LAB performing poorly. The LAB is the main heavyweight in the OADN, the FRB plays a supporting role, stepping up when the LAB is ineffective.

The greatest weakness of the OADN is its dependence on the SAN. The two branches are trained according to a combined loss function given by

$$L = \lambda L_{LAB} + (1 - \lambda) L_{FRB}$$

where λ determines the relative importance of each branch.

An important distinction between OADN and the gACNN family is that the gACNN family learns to detect which parts are occluded, while the OADN relies completely on the SAN's assessment on occlusion. It can be said that the largest weakness of OADN is its dependence on SAN [22].

The architecture of OADN is shown in figure 3.

D. WGAN for OFER

At the very start, all the images are preprocessed using MTCNN [34]. This network detects the face, the centres of the eyes and aligns the face so that the eye centres are horizontal. The face region is cropped and resized to be of the size 128x128 and the values of the pixel are normalized to 0 to 1 [23] .

This model proposes a variation of WGAN [35] which itself is a variation of the GAN network [36]. The model consists of 3 networks. A generator and 2 discriminator. The generator is a convolutional autoencoder [38] which takes input and output of the size 3x128x128. The input is an occluded image and the output is an reconstructed image without occlusion. A huge shortcoming of WGAN for OFER is that it is trained on synthetic occlusions. In the encoder, the depth of the image increases until it is eventually 512 layers deep. Each convolution uses a 3x3 kernels and has stride 1. Every convolution is followed by batch normalization and ReLU [37]. The output of the decoder is 512x8x8. The bottleneck consists of 2 fully connected layers with 1024 neurons where the incoming feature map is flattened and compressed into a 1024D vector. The decoder has a similar structure as the encoder but in the reverse direction and uses transpose convolutions [23] .

Both of the discriminators are based on VGG-19 and have the same architecture. The only real difference between them is in their fully connected layers. Both use 3x3 kernels, stride 1 and ReLU activation. After every convolution, the feature maps are downsized by pooling. Though it is not mentioned which type of pooling is used, it is very likely that it is max pooling considering that VGG-19 uses it. The pooling layers uses 2x2 filters with stride 2. After all this, both the models share 3 fully connected layers. First 2 of those contain 4096 neurons and the third one contains 1000 neurons [23].

Now coming to the difference between the generators. D1 is the real/fake discriminator. After the FC layers, a sigmoid function gives an output which tells how real or fake it is. An important thing to note here is that if the model is using Wasserstein distance, it shouldn't really be using a sigmoid function. D1's objective is to push the generator to make more realistic non-occluded images [23].

D2 is the expression classifier which after the FC layers sends the input through a softmax function and outputs

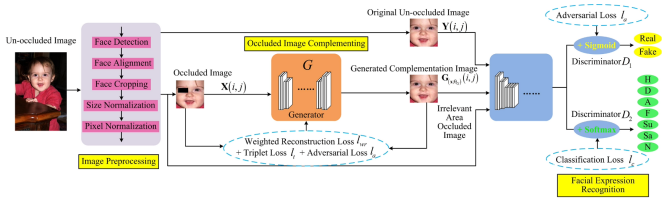

Fig. 4: Architecture for WGAN for OFER [23]

probabilities over 7 expressions: happiness, sadness, surprise, disgust, fear, anger, neutral [23].

Both discriminators are trained on 3 types of inputs, namely generated de-occluded images, original un-occluded images and irrelevant-area occluded images. Irrelevant area occluded images are images where the occlusion is on areas which do not carry a lot of information relating to facial expressions. This is done so that D2 learns authentic expression features since reconstructed images will carry subtle differences from natural images [23].

The model is trained using 4 loss functions. The first is weighted reconstruction loss. This measures pixel level similarity between generated and original non-occluded images. The second loss function is triplet loss. To calculate this, we define 3 terms. d_1 which is the difference between the generated and original image in the occluded area and d_2 which is the difference between the generated and occluded images in the occluded area. The objective is to make d_1 small and d_2 large. Another factor β is introduced to make sure that d_2 is not marginally larger than d_1 . The third loss function is adversarial loss from the WGAN. The final loss function is the classification loss which is the standard cross entropy loss on D2 [23].

The discriminator and generator are trained alternatively . For every time the discriminators are trained, the generator is trained 5 times.

The architecture of WGAN for OFER is given in figure 4

E. WLS-RF

Feature extraction for this model is done using SDM, or supervised descent method [39]. The model uses 3 features in total, of which 2 are derived from the landmarks detected by SDM, the third represents the edges, which in this model are detected by histogram of oriented gradients [40].

The model then trains random forests on the image, but not the entire image. A triangular mesh is defined using the facial landmarks and a mask is generated by randomly selecting connected triangles. This mask is allowed to grow until it covers $\frac{1}{10}^{th}$ of the original image [24].

The model detects occlusion using a hierarchical autoencoder which assigns a confidence score to each triangle. The autoencoder is trained on non-occluded data however, with 25% masking noise. The final prediction is simply the prediction of each triangle weighted by its confidence [24].

This model was also shown to be capable of detecting facial action units [41].

The architecture of WLS-RF is given in figure 5.

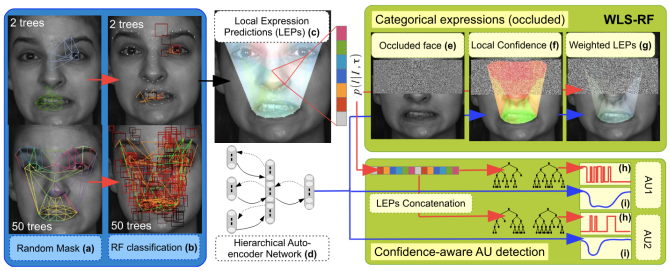

Fig. 5: Architecture for WLS-RF [24]

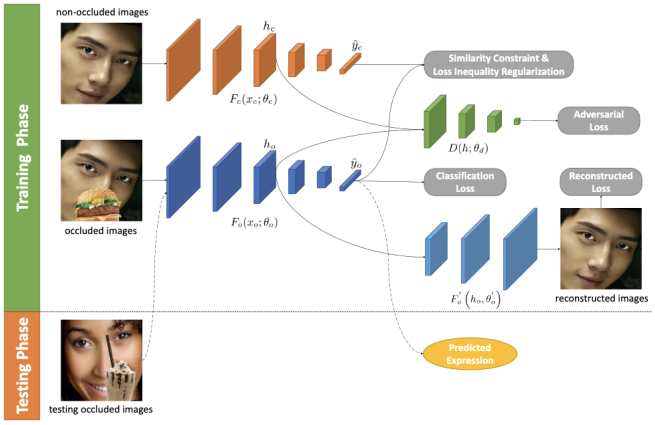

Fig. 6: Architecture of PI-CNN [25]

F. PI-CNN

This model deal with occlusion by treating the non-occluded images as privileged information [42]. The model consists of 2 CNNs, one classifies occluded facial expressions, this is our primary classifier and the other classifies non-occluded facial expressions. Both are pre-trained using cross entropy loss, after which the weights of the non-occluded network are fixed. Now, the occluded network is fine tuned with guidance in label space and feature space [25].

For label space, the two CNNs are subject to two constraints, the first is a similarity constraint which ensures that the output of both aren't wildly different and the second is loss inequality regularization [43]. The intuition behind it that the privileged information will always be more descriptive than than the non-privileged information, hence the output of the privileged network must always be closer to the base truth [25].

For guidance in feature space, a GAN-like network is introduced to ensure that the occluded network learns robust patterns that help in classification. The GAN-like network is only used in training and is only used to ensure that the classifier learns the desired kind of information, it is not used during inference [25].

The architecture of PI-CNN is shown in figure 6

G. RGBT

The first step in this model is detecting the face using the Viola-Jones face detector [44]. Then a gabor filter [46] bank is applied giving us a series of gabor images. Then, templates are extracted using a Monte Carlo algorithm [45].

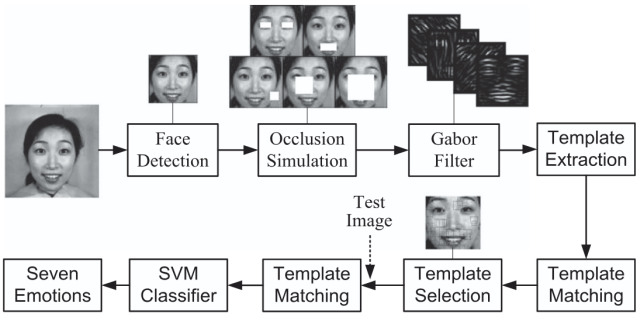

Fig. 7: Framework of RGBT [26]

Then an image from a certain class is selected and a random position is chosen, on this, one of the scales from the gabor filter bank is applied randomly. This is done to 4 different sizes of templates. This mechanism is introduced to deal with occlusion. Intuitively, occlusion will only cover a small area of the face, so if you randomly sample many such filters, you are bound to cover the entire face, when there is and isn't occlusion [26].

The next step is template matching, For the image, every template is applied, but not directly. It searches within a region twice the width and height of the template centered on the original template's position. Within this area, the template is slid across all positions and the distance is computed using one of two sparse metrics. By this point, we have a 28000 dimensional feature vector for each image, which will eventually have to be reduced [26].

We now come to training our classifier. A one against all SVM [47] is trained for each emotion and then pruning starts to reduce dimensionality. The importance of each template is calculated from the weights of the SVM and in the first iteration of training, the bottom half of templates are removed. This is done until the number is less than 1000, after which the templates are reduced 50 per iteration until the number goes below 100, after which they are reduced 10 per iteration. We should expect to get between 10 to 20 final templates [26].

In final classification, the emotion corresponding to the emotion with the highest decision value is the prediction [26].

The framework of RGBT is given in figure 7.

H. CT-DBN

This model uses a dual branch approach to deal with occlusion as seen earlier. Here we have a CNN and a Swin Transformer [48] which is a variation of the Transformer architecture [49].

First, the raw image goes through face detection using the Dlib library. Data augmentation is applied during training where the image is randomly flipped horizontally and color jitter is applied. From this image, local binary pattern (LBP) is applied to encode local texture information [27].

At this point, the model splits into 2 branches, first is the local feature extraction branch (LFEB) and the other is the global feature extraction and feature fusion branch (GFE-FFB). Both branches simultaneously process their inputs [27].

In the LFEB branch, the LBP image is fed into the first 5 layers of ResNet-18. This is more or less a standard CNN which outputs a prediction $Pred_1$. This is not used for the final classification, it is only to train the LFEB [27].

The GFE-FFB is a standard Swin Transformer along with all the accompanying embeddings and reshaping. The Swin Transformer consists of 2 sub-blocks, both are processed consecutively. The first sub-block is regular window multi-head self-attention (WMSA) and the second sub-block is shifted window multi-head self-attention (SWMSA) [27]

Then, global and local feature weights are computed, which are then used to compute a combined feature weight. This feature weight is used to calculate hybrid feature maps by using the global and local feature maps that were produced by both the branches [27].

The hybrid feature maps are passed through a fully connected layer to get $Pred_2$ which is our final prediction [27].

The architecture of CT-DBN is given in figure 8.

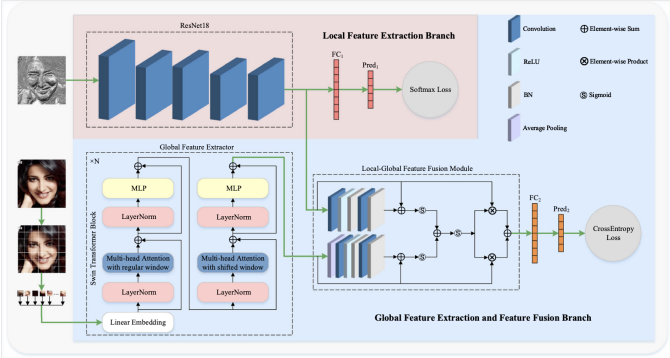

Fig. 8: Architecture of CT-DBN [27]

V. Detailed Analysis of the Proposed Models

Each of the models proposed is trained and tested on different datasets. Table 1 shows all the different datasets that the models used as either testing or validation. Furthermore, a lot of these models train on custom datasets constructed from images from two or more of the above datasets. This is the greatest limitation of this review. Due to constraints on time and competence, these models could not be tested on a level playing field. We will be relying completely on the figures provided by the original authors.

A factor that must be noted is that almost all of the above datasets were originally created as simple, non-occluded datasets. Occlusion-FERPlus contains real occlusions. It was collected by reviewing images in FERPlus and selecting those with genuine occlusions. Occlusion-FERPlus, as a result contains a relatively small dataset of 605 samples [28].

The only dataset that was collected from the start as a occlusion dataset is FED-RO [12]. This was collected by mining Bing and Google search with search queries like "smile+face+glasses". Then human annotators labelled the images. Then, images that overlapped with RAF-DB and AffectNet were removed. This too is a relatively small dataset of around 400 samples. The test results on this dataset are perhaps the closest to real world results we will get. This shows that there is still large scope for improvement since all the models who tested on FED-RO reported sub-75% accuracy.

For purposes of our comparison, we will be using the figures of the models for the RAF-DB [9] and CK+ [6] datasets. This is not a perfect way to compare as we are still comparing on two different datasets, but using the models performance relative to each other, we can get a rough idea of how the different methods fare.

Below is the accuracy of various models while testing with RAF-DB dataset

TABLE II: Accuracy of models on the RAF-DB dataset.

Model	Accuracy (%)
PG-CNN†	83.27
pACNN	78.05
gACNN†	85.07
OADN	87.16
WGAN for OFER	78.35
PI-CNN	81.97
CT-DBN	**88.40**

† Results reported in [22].

TABLE III: Accuracy of models on the CK+ dataset.

Model	Accuracy (%)
PG-CNN†	79.49
pACNN†	79.49
gACNN†	79.49
WLS-RF	**94.30**
PI-CNN	79.21
CT-DBN	81.50

† Trained on RAF-DB (cross-database evaluation).

Note the absence of RGBT [26] in both of the above tables. This is because there is no single figure provided by the

TABLE I: Evaluation datasets used by each model.

Model	RAF-DB	AffectNet	CK+	FED-RO	Occlusion-FERPlus[28]	FERPlus	BU4D[51]	SFEW	JAFFE
PG-CNN	✓	✓	✓	×	×	×	×	×	×
pACNN / gACNN	✓	✓	✓	✓	×	×	×	×	×
OADN	✓	✓	×	✓	✓	×	×	×	×
WGAN for OFER	✓	✓	×	×	×	×	×	×	×
WLS-RF	×	×	✓	×	×	×	✓	✓	×
PI-CNN	✓	✓	✓	✓	×	×	×	×	×
RGBT	×	×	✓	×	×	×	×	×	✓
CT-DBN	✓	×	✓	×	×	✓	×	×	×

authors. They provided results for their model only on specific types and sizes of occlusion.

Another important thing to keep in mind is that unless specified otherwise, most of the above models were trained on a combination of several datasets, often times including RAF-DB and CK+. Both of which were not collected with occlusion in mind.

This issue gets highlighted when models are tested on datasets like FED-RO [12] and Occlusion-FERPlus [28]. Table 4 shows the accuracy of the models that were tested on the occlusion specific datasets.

TABLE IV: Accuracy of models on occlusion datasets.

Model	FED-RO (%)	Occ-FERPlus (%)
pACNN	64.22	–
gACNN	65.27	–
OADN	68.11	84.57
PI-CNN	**69.75**	–

Out of all the models compared, we can see that the model which tried to reconstruct the image, WGAN for OFER [23] and PI-CNN [25], performed worse on CK+ and RAF-DB datasets than models which explicitly detected occlusion and dealt with it on a patch-wise basis.

It is certainly interesting to note that PI-CNN [25], a "reconstructivist" model scored higher accuracy than OADN [22], a patch based model, even though it scored higher while testing on the RAF-DB dataset. This suggests that privileged information could be more effective for real world facial recognition. Unfortunately, only one model was reviewed which used this principle and as of yet nothing definitively can be said.

The high accuracy for CT-DBN [27] and OADN [22] suggest that multiple branched models that base their inference on both local and global features fare better than those who do not.

The case of WLS-RF [24] seems to contradict the above statement, but it can be explained by realizing the fact that the model trains multiple overlapping random forests. Though this does not directly learn global features, it makes sure that the model learns relations between regions that may be spatially far away from each other. Despite this, WLS-RF [24] does not consider global features. It learns the relationship between features that are far away from each other but this is not the same as looking at the whole picture.

A final point to note is that CT-DBN [27] has no explicit occlusion handling mechanism. Its architecture is occlusion resistant to a large degree on its own.

VI. Conclusion

A. Summary of findings

1) This review can state that models that consider both local and global features fare better against occlusion than models that just account for local features, at least on synthetic occlusion testing sets.
2) A massive weakness of models that rely on facial landmarks is the landmark detection method they choose to use. This means that architectural decisions on landmark detection models/algorithms has a downstream effect, sometimes limiting on the facial expression recognition model. This is a large topic and warrants a closer look in a separate review.
3) Based on the figures available, there is no model that performs satisfactorily on OFER in real world conditions, as seen in the low accuracies scored by pACNN [12], gACNN [12], OADN [22] and PI-CNN [25] on the FED-RO dataset. While models have gotten reasonably good at detecting synthetic occlusions.

B. Limitations

The largest and limitation in this field is the unavailability of a large, high quality facial images datasets that has occlusions. There exists an opportunity for individual researchers to collect such a dataset. This alone can help massively in OFER.

C. Future Directions of Interest

1) This is in no way a comprehensive review. There are several models, like those proposed in [28][52][53][54][55] have not been covered in this review. Further, this review has not compared models on a level playing field, which has hindered the analysis presented.
2) As stated prior, it will be a huge contribution to collect a dataset consisting only of naturally occuring occluded faces.
3) There is still a need to propose a model that can reliably detect facial expressions in real world scenarios. No model scored a high accuracy on occlusion specific datasets.
4) Through this review, we can identify that networks that account for both local and global features fare better than those who do not. This tells us that any future models, which aim to achieve high accuracy in OFER are likely to account for global and local features.
5) The relatively good performance of PI-CNN [25] on the FED-RO [12] dataset shows us that models using privileged information could potentially work better in real world scenarios. Unfortunately, there are not many models using privileged information to deal with OFER. This could be a future area of academic interest.
6) The high accuracy scored by WLS-RF [24] on the CK+ [6] dataset shows that the approach of training multiple classifiers on overlapping regions could help in the creation of a high accuracy OFER model. Especially when you consider that WLS-RF [24] does not account for global features.
7) The relatively high accuracy scored by CT-DBN [27] when it doesn't even have a specific occlusion-handling mechanism, shows that a building on this architecture, by adding some occlusion handling mechanism could make it even better of a classifier. But this depends on occlusion handling mechanisms to be good enough to give it the necessary boost.

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